

Analysis of the Wave Equation in Non-Cartesian Coordinates

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January 2026

Abstract

This note discusses the theoretical foundations of tensor analysis and index notation required to maintain the coordinate invariance of physical laws in non-Cartesian coordinate systems. We first introduce the Einstein summation convention and define the geometric distinctions between covariant and contravariant basis vectors, as well as the role of the metric tensor in general curvilinear coordinates. Furthermore, the concepts of Christoffel symbols and covariant differentiation are established to account for the spatial variation of basis vectors. Based on this theoretical framework, we present a step-by-step formulation of the wave equation adapted for non-Cartesian coordinates. Specifically, the divergence term of the vector field is derived via the contraction of the covariant derivative to incorporate basis curvature, while the gradient term of the scalar field is modified using the metric tensor to ensure tensorial consistency of indices. This derivation provides a rigorous governing equation suitable for numerical computation in complex geometries.

1 Index Notation and Tensor Analysis

In physics, the formulation of physical laws must be independent of the choice of coordinate system made by the observer. Whether one adopts Cartesian coordinates (x, y, z) , spherical coordinates (r, θ, ϕ) , or any other admissible coordinate system, measurable physical quantities—such as energy, force, and distance—must represent the same physical reality.

This requirement of coordinate invariance naturally leads to the use of tensors. A tensor is not merely a collection of components but a geometric object whose defining property is its well-defined transformation behavior under changes of coordinates. Scalars, vectors, and second-order matrices commonly encountered in physics are, in fact, tensors of rank 0, 1, and 2, respectively. More generally, in three-dimensional space, a tensor of rank n requires 3^n components to represent its coordinate-dependent description.

Tensorial formulations, while essential for expressing physical laws in a coordinate-independent manner, often involve intricate algebraic manipulations. In particular, the repeated summations over multiple indices can make explicit expressions lengthy and obscure the underlying physical structure. To manage this complexity and to present tensor equations in a compact and transparent form, index notation is introduced and systematically employed. Index notation is a mathematical language that concisely describes complex vector and tensor operations by decomposing them into individual components.

1.1 Einstein Summation Convention

The most prominent feature of this notation is the use of the **Einstein summation convention**. The Einstein summation convention is a rule stating that when an index is repeated twice within a single term—once as an upper index and once as a lower index—a summation over that index is implied. This convention allows expressions to be written without explicitly using the summation symbol $\sum$, thereby greatly simplifying the notation. For example, dot product between two vectors can be expressed as followed:

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i = a_i b_i \tag{1}$$

In this notation, indices are broadly classified into two types: **dummy indices**, which are repeated within a single term, and **free indices**, which appear only once in the entire expression. Dummy indices are used solely to denote summation and therefore carry no intrinsic physical meaning, whereas free indices represent the components that remain in the expression. Consequently, free indices must appear with the same symbol and in the same position (upper or lower) on both sides of an equality.

1.2 Superscript and subscript index: Covariant and Contravariant

In a general coordinate system, two types of basis vectors exist.

Covariant basis vectors Covariant basis vectors are tangent to the coordinate curves and are obtained by partially differentiating the position vector $\mathbf{r}$ with respect to the coordinate variables x^i :

$$\mathbf{e}_i = \frac{\partial \mathbf{r}}{\partial x^i}.$$

Contravariant basis vectors Contravariant basis vectors are normal to the coordinate surfaces and are defined as the gradients of the scalar coordinate fields x^i :

$$\mathbf{e}^i = \nabla x^i.$$

From a geometric perspective, the covariant basis vectors indicate the direction in which the position changes when only x^i is varied while all other coordinates x^j ($j \neq i$) are held fixed, corresponding to the conventional notion of coordinate axes. In contrast, the contravariant basis vectors point in the direction normal to the surface $x^i = \text{constant}$ (an isosurface), along which the corresponding coordinate value changes most rapidly.

In a cartesian coordinate system, the covariant and contravariant basis vectors always have the same directions and magnitudes due to the orthogonality and normalization of the coordinate axes, and thus no distinction is necessary. However, in a general coordinate system, they represent different geometric entities, making it essential to clearly distinguish between them.

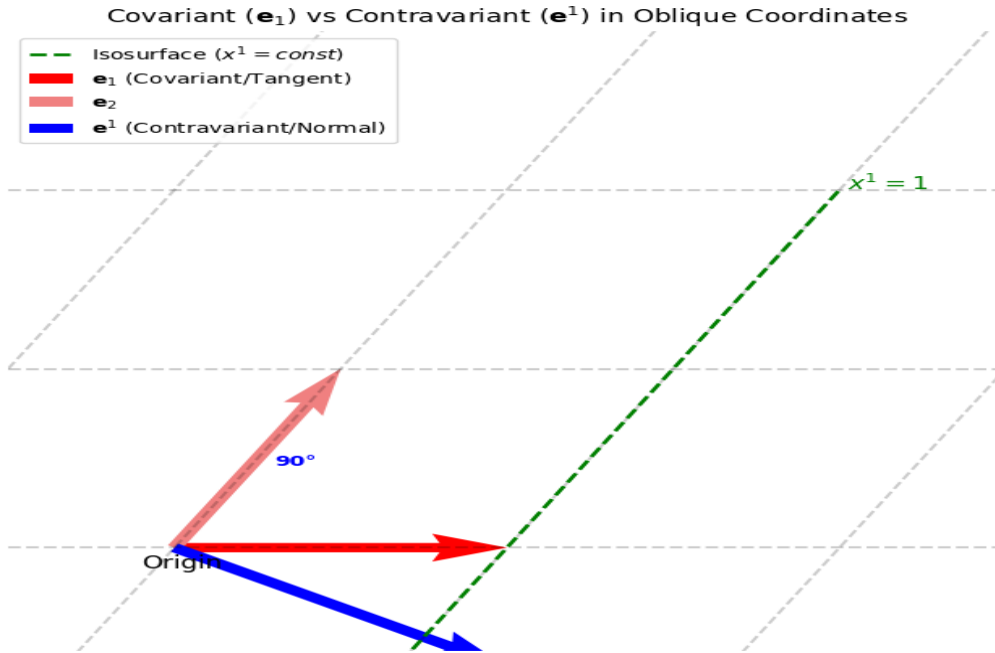


Figure 1: Covariant and Contravariant basis vectors in oblique coordinates. The covariant basis vector $\mathbf{e}_1$ is tangent to the coordinate curve, whereas the contravariant basis vector $\mathbf{e}^1$ is normal to the isosurface $x^1 = \text{const.}$

From Fig. 1, it can be observed that the covariant vector $\mathbf{e}_1$ along the axis direction and the contravariant vector $\mathbf{e}^1$, which is normal to the corresponding isosurface, form an angle of 30° and are clearly distinct. This illustrates that, in a non-Cartesian coordinate system, the covariant and contravariant basis vectors do not generally coincide.

Although it is known that these two bases do not coincide in a non-Cartesian coordinate system, this fact alone does not explain why they must necessarily be distinguished. The reason is that the two vectors serve fundamentally different roles within the coordinate system.

The covariant basis vectors are used to represent vectors in the coordinate space. The components associated with each basis are referred to as **contravariant components**, and for an arbitrary vector $\mathbf{v}$, this can be expressed as follows.

$$\mathbf{v} = v^i e_i \quad (2)$$

Conversely, the contravariant basis vectors are used to extract the components of a vector along specific directions. Likewise, for an arbitrary vector $\mathbf{v}$, this can be expressed as follows.

$$v^i = \mathbf{v} \cdot \mathbf{e}^i \quad (3)$$

Thus, owing to the differences in their geometric meanings and roles in operations within the coordinate space, these two types of basis vectors must be treated and distinguished rigorously in analytical procedures.

1.3 Metric Tensor

Then, how can the relationship between covariant and contravariant vectors be described? To address this question, we must first examine the metric tensor. The metric tensor is a tensor that encapsulates all the geometric information of a coordinate system—such as the angles between coordinate axes and the scale factors along each axis—and is defined as follows.

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j \quad (4)$$

Examining each term in this expression, g_{11} is the inner product of $\mathbf{e}_1$ with itself and is therefore related to the scale (or metric coefficient) of the x^1 -axis, while g_{12} is the inner product of $\mathbf{e}_1$ and $\mathbf{e}_2$, and is thus associated with the angle between the x^1 - and x^2 -axes. In this way, each entry of the metric tensor represents specific geometric information of the coordinate system.

One of the key roles of the metric tensor is that it is used to define distances in a coordinate system. In Cartesian coordinates, an infinitesimal displacement ds^2 can be expressed using the Pythagorean theorem as the sum of the squares of the infinitesimal displacements along each coordinate axis. However, in non-Cartesian coordinates, the covariant and contravariant bases are distinct, and motion along one coordinate direction generally influences the components along other directions. As a result, the distance cannot be represented simply as a sum of squared components; instead, it can be expressed using the metric tensor as follows.

$$ds^2 = g_{ij} dx^i dx^j \quad (5)$$

As can be seen from this expression, the metric tensor g_{ij} compensates for the curvature of the space and thereby ensures the preservation of physical quantities such as length.

Then, how can the relationship between the contravariant and covariant basis vectors be expressed using the metric tensor? To determine this, we must first examine the fundamental relationship that exists between the covariant and contravariant basis vectors.

$$\mathbf{e}^j \cdot \mathbf{e}_i = \delta_i^j \quad (6)$$

For $i \neq j$, the basis vector of a given coordinate direction must be orthogonal to the normal vector of a different coordinate surface, and thus their inner product must vanish. For $i = j$, the normalization of

the coordinate system implies that the inner product equals unity. Accordingly, this relationship can be expressed using the Kronecker delta, as shown in the above equation. This expression demonstrates the duality between the covariant and contravariant basis vectors.

From this expression, let us derive the relationship between the covariant and contravariant basis vectors. We seek to express the contravariant basis vector e^i as a linear combination of the covariant basis vectors e_k . Since both sets of vectors span the same space, we can write:

$$e^i = A^{ik} e_k \quad (7)$$

where A^{ik} represents the undetermined coefficients. To determine these coefficients, we take the inner product (dot product) of both sides with the covariant basis vector e_j .

Left-Hand Side (LHS): Using the duality condition (orthonormality) of the basis vectors:

$$\text{LHS} = e^i \cdot e_j = \delta_j^i \quad (8)$$

Right-Hand Side (RHS): Using the linearity of the dot product and the definition of the metric tensor ($g_{kj} = e_k \cdot e_j$):

$$\text{RHS} = (A^{ik} e_k) \cdot e_j = A^{ik} (e_k \cdot e_j) = A^{ik} g_{kj} \quad (9)$$

Comparison and Solution: Equating the LHS and RHS yields:

$$A^{ik} g_{kj} = \delta_j^i \quad (10)$$

In the context of linear algebra, this equation implies that the matrix formed by the coefficients A is the inverse of the metric tensor matrix g . Therefore, we identify:

$$A^{ik} = (g^{-1})^{ik} = g^{ik} \quad (11)$$

Substituting this result back into Equation (11), we obtain the final relationship for raising indices:

$$e^i = g^{ik} e_k \quad (12)$$

1.4 Covariant Derivative

Another fundamental consequence arising from the distinction between covariant and contravariant bases in a non-Cartesian coordinate system is the concept of **covariant differentiation**.

Differentiation is the tool used to analyze how fields change. However, in tensor calculus, simple partial differentiation is often insufficient because the coordinate system itself may change (curve) from point to point. The Partial derivative of tensor is defined as followed.

$$\partial_k A_i = \frac{\partial A_i}{\partial x^k} \quad (13)$$

In Cartesian coordinates, the basis vectors are constant in space, and therefore only the components of a tensor vary with position. As a result, the partial derivative of a tensor with respect to the Cartesian coordinates transforms as a tensor. However, in curvilinear coordinates, the basis vectors generally depend on position and vary as one moves through space. As a consequence, they may rotate and change their magnitude. A naive application of the partial derivative accounts only for the variation of the tensor components and neglects the spatial variation of the basis vectors. This omission causes the resulting quantity to fail to transform as a tensor. Then, why must we account for the variation of basis vectors under coordinate displacements? To address this, let us take the partial derivative of an arbitrary vector $\mathbf{v}$ with respect to a specific coordinate direction x^k . An arbitrary vector $\mathbf{v}$ can be represented in the given coordinate system as a combination of its contravariant components v^i and

the covariant basis vectors $\mathbf{e}_i$. Using this representation, the partial derivative with respect to x^k can be expanded by applying the product rule, yielding the following expression.

$$\frac{\partial v}{\partial x^k} = \frac{\partial}{\partial x^k} (v^i e_i) = \underbrace{\left(\frac{\partial v^i}{\partial x^k} \right) e_i}_{\text{change of variable}} + \underbrace{v^i \left(\frac{\partial e_i}{\partial x^k} \right)}_{\text{change of basis}} \quad (14)$$

Therefore, in curvilinear coordinates, the partial derivative of a tensor field generates additional terms due to the position dependence of the basis vectors, and these terms must be properly accounted for in order to preserve tensorial character.

To fix the issue of rotating basis vectors, we introduce the **Covariant Derivative**. It accounts for both the change in the component (A_i) and the change in the basis vectors. For a vector A_i , the covariant derivative is:

$$\nabla_k A^i = \partial_k A^i + \Gamma_{mk}^i A^m \quad (15)$$

- $\partial_k A^i$: The rate of change of the component (Partial Derivative).
- $\Gamma_{mk}^i A^m$: The correction term accounting for the curvature of the coordinate system.
- Γ_{mk}^i : **Christoffel Symbols** (Not a tensor itself, but a correction factor).

Based on the physical interpretation of the Christoffel symbols, they are defined by the following expression.

$$\frac{\partial e_i}{\partial x^j} = \Gamma_{ij}^k e_k \quad (16)$$

Since the derivative of the basis vector e_i with respect to x^j is itself a vector in the coordinate space, it must be expressible as a linear combination of the covariant basis vectors. The quantities that provide this mapping are the Christoffel symbols.

2 Formulation of the Wave Equation in Non-Cartesian Coordinates

Let us describe the wave equation formulated in a coordinate space. When the equation is expressed in a form that is convenient for numerical computation on a computer, it can be written as follows.

$$\nabla \cdot (a \nabla u) = \frac{1}{b} \frac{\partial^2 u}{\partial t^2} \quad (17)$$

For computational convenience, let us introduce an auxiliary variable $\mathbf{v}(\mathbf{x}, t)$ and decompose the second-order partial differential equation into two first-order partial differential equations.

$$\frac{\partial u}{\partial t} = b \nabla \cdot \mathbf{v} \quad (18)$$

$$\frac{\partial \mathbf{v}}{\partial t} = a \nabla u \quad (19)$$

To analyze this equation in a non-Cartesian coordinate system, each equation originally formulated in Cartesian coordinates must be transformed to be consistent with the non-Cartesian coordinate system.

First, let us examine Equation (18). This equation is defined in terms of the divergence of the vector quantity $\mathbf{v}$. In Cartesian coordinates, the divergence is given by $\nabla \cdot \mathbf{v} = \partial_i v_i$; however, in a non-Cartesian coordinate system, this definition cannot be used directly. As mentioned earlier, in a curvilinear coordinate system, the coordinate axes themselves must be taken into account during differentiation. Therefore, the expression must be formulated using the covariant derivative rather than the ordinary partial derivative. Accordingly, the equation is modified as follows.

Step 1: Expansion of Operators and Vectors First, express the del operator ∇ and the vector $\mathbf{v}$ using the basis vectors $\mathbf{e}^i$ and $\mathbf{e}_j$. Note that Einstein summation convention is used.

$$\nabla \cdot \mathbf{v} = (\mathbf{e}^i \partial_i) \cdot (v^j \mathbf{e}_j) \quad (20)$$

Step 2: Application of the Product Rule Since the basis vectors $\mathbf{e}_j$ in curvilinear coordinates are functions of position, the partial derivative ∂_i applies to both the component v^j and the basis vector $\mathbf{e}_j$ according to the product rule.

$$\nabla \cdot \mathbf{v} = \mathbf{e}^i \cdot [(\partial_i v^j) \mathbf{e}_j + v^j (\partial_i \mathbf{e}_j)] \quad (21)$$

This expands into two distinct terms:

$$\nabla \cdot \mathbf{v} = \underbrace{(\partial_i v^j)(\mathbf{e}^i \cdot \mathbf{e}_j)}_{\text{Term I}} + \underbrace{v^j \mathbf{e}^i \cdot (\partial_i \mathbf{e}_j)}_{\text{Term II}} \quad (22)$$

Step 3: Simplification of Term I (Scalar Component) Using the orthogonality property of the basis vectors, $\mathbf{e}^i \cdot \mathbf{e}_j = \delta_j^i$ (Kronecker delta), the first term simplifies to the standard divergence form:

$$\text{Term I} = (\partial_i v^j) \delta_j^i = \partial_i v^i \quad (23)$$

Step 4: Evaluation of Term II (Christoffel Symbols) For the second term, we substitute the derivative of the basis vector with the Christoffel symbols of the second kind, defined as $\partial_i \mathbf{e}_j = \Gamma_{ij}^k \mathbf{e}_k$.

$$\text{Term II} = v^j \mathbf{e}^i \cdot (\Gamma_{ij}^k \mathbf{e}_k) = v^j \Gamma_{ij}^k (\mathbf{e}^i \cdot \mathbf{e}_k) \quad (24)$$

Since $\mathbf{e}^i \cdot \mathbf{e}_k = \delta_k^i$, the indices contract ($k \rightarrow i$):

$$\text{Term II} = v^j \Gamma_{ij}^k \delta_k^i = v^j \Gamma_{ij}^i \quad (25)$$

Step 5: Final Expression Combining the results from Step 3 and Step 4, we obtain the full expression for divergence. This is equivalent to the contraction of the covariant derivative $\nabla_i v^i$.

$$\nabla \cdot \mathbf{v} = \partial_i v^i + \Gamma_{ij}^i v^j \equiv \nabla_i v^i \quad (26)$$

Equation (26) shows that the divergence consists of the partial derivative of the components plus a correction term involving the Christoffel symbols, which accounts for the curvature of the coordinate system. Finally, the total equation changes as follows.

$$\partial_t u = b \nabla_i \cdot v^i = b(\partial_i v^i + \Gamma_{mi}^i v^m) \quad (27)$$

Next, let us examine Equation (19). When the equation is interpreted in a non-Cartesian coordinate system, the left-hand side corresponds to the time derivative of the contravariant vector $\mathbf{v}$, and therefore consists of contravariant components, whereas the right-hand side represents the gradient of the scalar field u , which yields covariant components. In other words, the indices on the left-hand side and the right-hand side are not consistent. Consequently, the gradient operator must be modified appropriately when formulated in a non-Cartesian coordinate system.

We aim to derive the component form of the governing equation by equating the time evolution of a vector field to a gradient-driven term. The derivation proceeds in four steps.

Step 1: Expansion of the Left-Hand Side (LHS)

The vector field $\mathbf{v}$ is defined as the linear combination of its contravariant components v^i and the covariant basis vectors $\mathbf{e}_i$:

$$\mathbf{v} = v^i \mathbf{e}_i \quad (28)$$

Differentiating with respect to time t , and assuming the coordinate system is time-independent (i.e., the basis vectors $\mathbf{e}_i$ are constant in time), we obtain:

$$\text{LHS} = \frac{\partial \mathbf{v}}{\partial t} = \frac{\partial v^i}{\partial t} \mathbf{e}_i \quad (29)$$

The result is expressed in terms of the **covariant basis** $\mathbf{e}_i$.

Step 2: Expansion of the Right-Hand Side (RHS)

The gradient ∇u is mathematically a 1-form (covector), which naturally couples with the **contravariant basis** $\mathbf{e}^j$:

$$\nabla u = \frac{\partial u}{\partial x^j} \mathbf{e}^j = (\partial_j u) \mathbf{e}^j \quad (30)$$

Thus, the right-hand side term is given by:

$$\text{RHS} = a(\partial_j u) \mathbf{e}^j \quad (31)$$

Here, the result is expressed in terms of the **contravariant basis** $\mathbf{e}^j$.

Step 3: Basis Unification using the Metric Tensor

To equate the LHS (Equation. 29) and the RHS, we must unify the basis vectors. We use the metric tensor to raise the index of the covariant basis, expressing $\mathbf{e}^j$ in terms of $\mathbf{e}_k$:

$$\mathbf{e}^j = g^{jk} \mathbf{e}_k \quad (32)$$

Substituting this relationship into the RHS:

$$\begin{aligned} \text{RHS} &= a(\partial_j u)(g^{jk} \mathbf{e}_k) \\ &= a(g^{jk} \partial_j u) \mathbf{e}_k \end{aligned} \quad (33)$$

Step 4: Coefficient Comparison

We now equate the LHS and the modified RHS. For clarity, we rename the dummy index k to i on the right-hand side:

$$\frac{\partial v^i}{\partial t} \mathbf{e}_i = (a g^{ji} \partial_j u) \mathbf{e}_i \quad (34)$$

Since the basis vectors $\mathbf{e}_i$ are linearly independent, the coefficients corresponding to each basis vector must be identical. Therefore, we arrive at the final component equation:

$$\frac{\partial v^i}{\partial t} = a g^{ji} \partial_j u \quad (35)$$